Impact-calibrated infit bands, validation

Upper side of the band from second-dimension misfit, against a same-item counterfactual

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Purpose

An effect-size floor judges an item's infit by an FWB interval for its population MSQ against a band $[1 - \delta_i^-, 1 + \delta_i^+]$. This study calibrates the upper side, δ_i^+, from the consequences of misfit for person scores, and checks that the calibration behaves. The design and the decisions behind it are in `infit-impact-delta-design.md`. The simulation is `infit_impact_validation.R`.

Design

- **Counterfactual.** For a target item i and correlation ρ, datasets

  (a) item i responds to θ_2, (b) item i responds to θ_1, and (c) item i removed share every person and every uniform draw. Impact of the misfit is (a) against (b), both judged against the true θ_1. (a) against (c) is the keep-or-remove comparison.
- **Group pair.** A random half of the sample is shifted by 0.5 SD on the item-specific factor of θ_2, with θ_1 unchanged, for metric E.
- **Analysis.** CML refit, WLE with SEM, linked to the generating scale through the items other than i. Sum scores alongside.
- **Designs.** `d20`: 20 dichotomous items evenly spaced on [−2, 2] (the Part A design), targets at about 0, −1 and −2 logits from the person mean. `phq9`: the package data, 9 items with 4 categories, targets nearest the person mean, at the median distance, and farthest away.
- **Persons.** Resampled WLE from a base dataset (2,000 respondents for d20, the complete phq9 cases), as in the `dgp = "resample"` cutoffs.
- **Grid.** ρ = .3, .5, .6, .7, .8, .85, .9, .95 and 1. $\rho = 1$ is the fitting item, where every impact must be zero.

- **Size.** n = 20,000 per dataset as a population proxy, 40 replicates per cell. For the on-target items only, also n = 2,000 and 5,000, to check whether the impact estimates move with n. All tables except the stability section use n = 20,000.

Impact metrics (section 5 of the design note):

	Metric	Units
A	RMSE of $\hat{\theta} - \theta_1$, (a) over (b)	ratio
B	mean $\mid \hat{\theta}_a - \hat{\theta}_b \mid / \mathrm{SEM}_b$, and the share above 1	SEM
C	mean KL between $N(\hat{\theta}_a, \mathrm{SEM}_a^2)$ and $N(\hat{\theta}_b, \mathrm{SEM}_b^2)$	nats
E	spurious group difference per SD of group difference on the item-specific factor	SD of θ_1 (WLE) or of the sum score
F	largest difference in conditional bias $E(\hat{\theta} - \theta_1 \mid \theta_1)$ between (a) and (b)	SEM at that θ_1

Checks

```
S |>
  group_by(design, target) |>
  summarise(cells = n_distinct(rho), reps = n(),
            impact_at_rho1 = max(abs(c(rmse_a[rho == 1] - rmse_b[rho == 1],
                                       shift_mean[rho == 1], kl_mean[rho == 1],
                                       E_wle[rho == 1]))),
            msq_at_rho1 = mean(msq[rho == 1]), .groups = "drop") |>
  kb(digits = 4)
```

design	target	cells	reps	impact_at_rho1	msq_at_rho1
d20	on	9	360	0	1.0010
d20	mid	9	360	0	1.0003
d20	off	9	360	0	1.0012
phq9	on	9	360	0	1.0002
phq9	mid	9	360	0	1.0016
phq9	off	9	360	0	1.0023

0 replicates failed. About 9 minutes on 10 cores. At $\rho = 1$ every impact must be exactly 0 and the MSQ near 1.

MSQ and impact by ρ

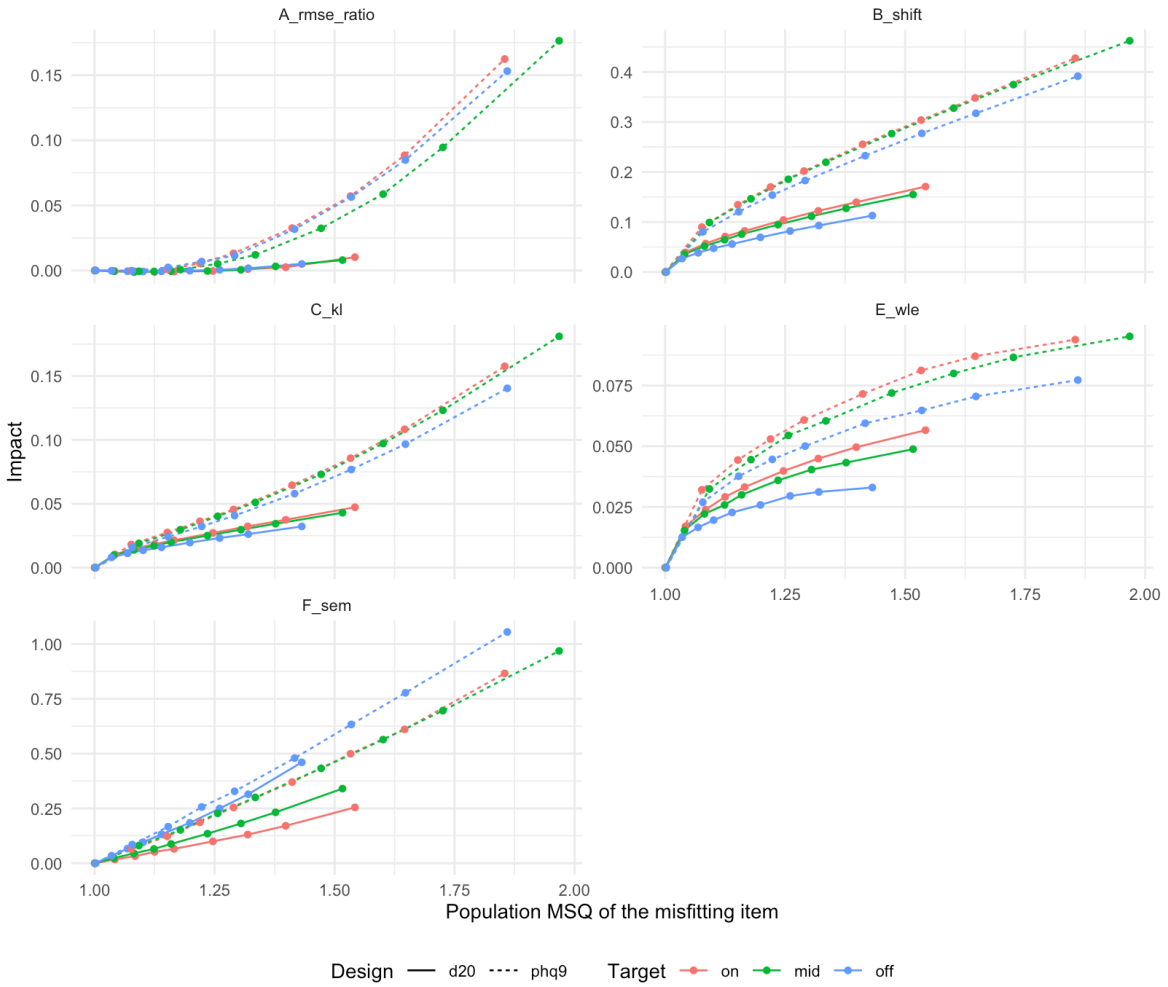
```
cell |>
  select(design, target, rho, msq, other_min, other_max, A_rmse_ratio, B_shift,
         B_gt1, C_kl, E_wle, E_sum, F_sem) |>
  arrange(design, target, desc(rho)) |>
  kb(digits = c(0, 0, 2, 3, 3, 3, 3, 3, 3, 4, 3, 3, 3))
```

design	target	rho	msq	other_min	other_max	A_rmse_ratio	B_shift	B_gt1	C_kl	E_wle	E_sum	F_sem
d20	on	1.00	1.001	0.984	1.016	1.000	0.000	0.000	0.0000	0.000	0.000	0.000
d20	on	0.95	1.042	0.982	1.013	0.999	0.039	0.000	0.0104	0.017	0.017	0.017
d20	on	0.90	1.084	0.980	1.011	0.999	0.057	0.000	0.0149	0.024	0.024	0.032
d20	on	0.85	1.125	0.978	1.008	0.999	0.071	0.000	0.0185	0.029	0.029	0.052
d20	on	0.80	1.166	0.975	1.006	0.999	0.082	0.000	0.0214	0.033	0.032	0.065
d20	on	0.70	1.246	0.971	1.001	1.000	0.104	0.000	0.0272	0.040	0.038	0.100
d20	on	0.60	1.319	0.965	0.997	1.001	0.122	0.000	0.0323	0.045	0.042	0.131
d20	on	0.50	1.398	0.962	0.991	1.003	0.140	0.000	0.0374	0.050	0.046	0.171
d20	on	0.30	1.542	0.953	0.982	1.010	0.171	0.000	0.0473	0.057	0.051	0.255
d20	mid	1.00	1.000	0.984	1.016	1.000	0.000	0.000	0.0000	0.000	0.000	0.000
d20	mid	0.95	1.040	0.982	1.015	0.999	0.036	0.000	0.0097	0.015	0.015	0.023
d20	mid	0.90	1.082	0.980	1.011	0.999	0.052	0.000	0.0138	0.022	0.021	0.045
d20	mid	0.85	1.124	0.978	1.010	0.999	0.065	0.000	0.0171	0.026	0.025	0.065
d20	mid	0.80	1.159	0.976	1.007	0.999	0.076	0.000	0.0200	0.030	0.028	0.088
d20	mid	0.70	1.235	0.970	1.004	1.000	0.095	0.000	0.0251	0.036	0.033	0.135
d20	mid	0.60	1.305	0.966	1.000	1.001	0.111	0.000	0.0297	0.040	0.037	0.181
d20	mid	0.50	1.377	0.960	0.997	1.003	0.127	0.000	0.0344	0.043	0.039	0.232
d20	mid	0.30	1.516	0.950	0.989	1.008	0.155	0.000	0.0430	0.049	0.043	0.340
d20	off	1.00	1.001	0.983	1.016	1.000	0.000	0.000	0.0000	0.000	0.000	0.000
d20	off	0.95	1.035	0.983	1.014	1.000	0.027	0.000	0.0080	0.013	0.011	0.034
d20	off	0.90	1.069	0.981	1.013	0.999	0.039	0.000	0.0112	0.017	0.014	0.067
d20	off	0.85	1.101	0.978	1.011	0.999	0.048	0.000	0.0136	0.020	0.017	0.096
d20	off	0.80	1.139	0.976	1.010	0.999	0.056	0.000	0.0159	0.023	0.019	0.132
d20	off	0.70	1.198	0.973	1.009	1.000	0.069	0.000	0.0196	0.026	0.022	0.184
d20	off	0.60	1.260	0.967	1.007	1.001	0.082	0.000	0.0231	0.030	0.025	0.250
d20	off	0.50	1.320	0.964	1.005	1.002	0.093	0.000	0.0263	0.031	0.027	0.315
d20	off	0.30	1.432	0.952	0.999	1.005	0.113	0.000	0.0322	0.033	0.028	0.460
phq9	on	1.00	1.000	0.986	1.014	1.000	0.000	0.000	0.0000	0.000	0.000	0.000
phq9	on	0.95	1.076	0.976	1.002	1.000	0.090	0.001	0.0181	0.032	0.033	0.065
phq9	on	0.90	1.151	0.967	0.992	1.001	0.135	0.001	0.0275	0.044	0.045	0.123
phq9	on	0.85	1.220	0.957	0.982	1.005	0.170	0.002	0.0363	0.053	0.054	0.187
phq9	on	0.80	1.289	0.947	0.974	1.013	0.202	0.003	0.0455	0.061	0.062	0.254
phq9	on	0.70	1.411	0.931	0.955	1.033	0.255	0.008	0.0645	0.072	0.073	0.370
phq9	on	0.60	1.533	0.914	0.939	1.057	0.304	0.018	0.0857	0.081	0.081	0.499
phq9	on	0.50	1.646	0.897	0.922	1.088	0.348	0.031	0.1083	0.087	0.088	0.611
phq9	on	0.30	1.854	0.866	0.894	1.162	0.428	0.065	0.1576	0.094	0.097	0.866

design	target	rho	msq	other_min	other_max	A_rmse_ratio	B_shift	B_gt1	C_kl	E_wle	E_sum	F_sem
phq9	mid	1.00	1.002	0.987	1.013	1.000	0.000	0.000	0.0000	0.000	0.000	0.000
phq9	mid	0.95	1.092	0.976	1.003	0.999	0.099	0.000	0.0190	0.032	0.035	0.080
phq9	mid	0.90	1.179	0.964	0.993	1.001	0.146	0.001	0.0297	0.044	0.048	0.152
phq9	mid	0.85	1.257	0.956	0.983	1.005	0.186	0.003	0.0402	0.054	0.057	0.228
phq9	mid	0.80	1.335	0.945	0.974	1.012	0.220	0.006	0.0511	0.060	0.064	0.300
phq9	mid	0.70	1.472	0.926	0.958	1.032	0.277	0.015	0.0730	0.072	0.076	0.433
phq9	mid	0.60	1.601	0.911	0.940	1.059	0.328	0.028	0.0972	0.080	0.084	0.564
phq9	mid	0.50	1.726	0.893	0.924	1.095	0.375	0.048	0.1232	0.087	0.090	0.696
phq9	mid	0.30	1.968	0.861	0.895	1.176	0.463	0.094	0.1811	0.095	0.100	0.968
phq9	off	1.00	1.002	0.985	1.015	1.000	0.000	0.000	0.0000	0.000	0.000	0.000
phq9	off	0.95	1.078	0.977	1.004	1.000	0.080	0.000	0.0158	0.027	0.027	0.085
phq9	off	0.90	1.153	0.967	0.998	1.002	0.121	0.001	0.0242	0.038	0.038	0.166
phq9	off	0.85	1.223	0.961	0.990	1.007	0.154	0.002	0.0323	0.045	0.044	0.256
phq9	off	0.80	1.292	0.950	0.981	1.011	0.183	0.003	0.0407	0.050	0.050	0.328
phq9	off	0.70	1.417	0.933	0.967	1.032	0.233	0.009	0.0579	0.059	0.059	0.479
phq9	off	0.60	1.535	0.919	0.954	1.056	0.277	0.017	0.0768	0.065	0.065	0.633
phq9	off	0.50	1.648	0.904	0.938	1.085	0.317	0.030	0.0967	0.070	0.070	0.778
phq9	off	0.30	1.860	0.876	0.915	1.153	0.392	0.065	0.1404	0.077	0.078	1.055

`other_min` and `other_max` are the lowest and highest MSQ among the other items in dataset (a): the cascade caused by one misfitting item.

```
cell |>
  select(design, target, msq, A_rmse_ratio, B_shift, C_kl, E_wle, F_sem) |>
  mutate(A_rmse_ratio = A_rmse_ratio - 1) |>
  pivot_longer(c(A_rmse_ratio, B_shift, C_kl, E_wle, F_sem),
               names_to = "metric") |>
  ggplot(aes(msq, value, colour = target, linetype = design)) +
  geom_line() + geom_point(size = 1.2) +
  facet_wrap(~ metric, scales = "free_y", ncol = 2) +
  labs(x = "Population MSQ of the misfitting item", y = "Impact",
       colour = "Target", linetype = "Design") +
  theme_minimal(base_size = 10) +
  theme(legend.position = "bottom")
```



A for this figure is the RMSE ratio minus 1.

Monotonicity. The calibration inverts impact as a function of MSQ, which needs impact to rise as MSQ rises. Rank correlation over the grid, per item and metric (1 means perfectly monotone):

```
cell |>
  group_by(design, target) |>
  summarise(across(c(A_rmse_ratio, B_shift, C_kl, E_wle, F_sem),
                   \(v) cor(msq, v, method = "spearman")),
            .groups = "drop") |>
  kb(digits = 2)
```

design	target	A_rmse_ratio	B_shift	C_kl	E_wle	F_sem
d20	on	0.60	1	1	1	1
d20	mid	0.70	1	1	1	1
d20	off	0.60	1	1	1	1
phq9	on	0.98	1	1	1	1
phq9	mid	0.98	1	1	1	1
phq9	off	0.98	1	1	1	1

Calibrated δ^{+}

For each item and metric, the MSQ at which impact reaches the tolerance, read by linear interpolation along the grid, minus 1. `>` means the tolerance was not reached within the grid (the largest MSQ reached is shown). The tolerances are illustrations of what a user might choose.

```
tol <- tribble(
  ~metric,        ~tau,  ~label,
  "A_rmse_ratio", 1.025, "A: RMSE +2.5%",
  "A_rmse_ratio", 1.05,  "A: RMSE +5%",
  "B_shift",      0.25,  "B: mean shift 0.25 SEM",
  "B_gt1",        0.05,  "B: 5% move > 1 SEM",
  "C_kl",         0.03,  "C: KL 0.03",
  "E_wle",        0.05,  "E: 0.05 SD per SD",
  "E_wle",        0.10,  "E: 0.10 SD per SD",
  "F_sem",        0.25,  "F: 0.25 SEM"
)
invert <- function(msq, v, tau) {
  o <- order(msq); msq <- msq[o]; v <- v[o]
  j <- which(v >= tau)[1]
  if (is.na(j)) return(sprintf(">%.2f", max(msq) - 1))
  if (j == 1) return(sprintf("%.2f", msq[1] - 1))
  x <- msq[j - 1] + (tau - v[j - 1]) * (msq[j] - msq[j - 1]) / (v[j] - v[j - 1])
  sprintf("%.2f", x - 1)
}
cell |>
  group_by(design, target, rel_loc) |>
  group_modify(function(d, key) {
    tibble(label = tol$label,
           delta = mapply(function(m, t) invert(d$msq, d[[m]], t),
                          tol$metric, tol$tau))
  }) |>
  ungroup() |>
  pivot_wider(names_from = label, values_from = delta) |>
  kb(digits = 2)
```

design	target	rel_loc	A: RMSE +2.5%	A: RMSE +5%	B: mean shift 0.25 SEM	B: 5% move > 1 SEM	C: KL 0.03	E: 0.05 SD per SD	E: 0.10 SD per SD	F: 0.25 SEM
d20	on	0.07	>0.54	>0.54	>0.54	>0.54	0.29	0.41	>0.54	0.53
d20	mid	0.98	>0.52	>0.52	>0.52	>0.52	0.31	>0.52	>0.52	0.40
d20	off	-2.06	>0.43	>0.43	>0.43	>0.43	0.39	>0.43	>0.43	0.26
phq9	on	-0.56	0.36	0.50	0.40	0.77	0.17	0.20	>0.85	0.28
phq9	mid	0.76	0.42	0.56	0.41	0.73	0.18	0.22	>0.97	0.28
phq9	off	-1.48	0.38	0.50	0.46	0.77	0.20	0.29	>0.86	0.22

Keep or remove

RMSE with the misfitting item kept, over RMSE with it removed (dataset c). Below 1, keeping the item measures θ_1 better despite the misfit. The table shows the ratio by ρ and the MSQ at which it crosses 1.

```
cell |>
  select(design, target, rho, msq, keep_vs_remove) |>
  pivot_wider(names_from = rho, values_from = c(msq, keep_vs_remove),
              names_glue = "{.value}_{rho}") |>
  select(design, target, starts_with("keep_vs_remove")) |>
  kb(digits = 3)
```

design	target	keep_vs_remove_0.3	keep_vs_remove_0.5	keep_vs_remove_0.6	keep_vs_remove_0.7	keep_vs_remove_0.8	keep_vs_remove_0.85	keep_vs_remove_0.9	keep_vs_remove_0.95	keep_vs_remove_1
d20	on	0.985	0.978	0.977	0.976	0.975	0.975	0.975	0.976	0.975
d20	mid	0.984	0.980	0.977	0.976	0.976	0.976	0.976	0.977	0.977
d20	off	0.986	0.982	0.982	0.982	0.980	0.980	0.981	0.981	0.981
phq9	on	1.108	1.037	1.007	0.983	0.964	0.958	0.953	0.953	0.952
phq9	mid	1.120	1.042	1.008	0.982	0.964	0.957	0.952	0.952	0.952
phq9	off	1.102	1.036	1.009	0.985	0.967	0.961	0.957	0.955	0.954

```
cell |>
  group_by(design, target) |>
  summarise(msq_where_equal = {
    o <- order(msq); m <- msq[o]; k <- keep_vs_remove[o]
    j <- which(k >= 1)[1]
    if (is.na(j)) NA_real_ else if (j == 1) m[1] else
      m[j - 1] + (1 - k[j - 1]) * (m[j] - m[j - 1]) / (k[j] - k[j - 1])
  }, .groups = "drop") |>
  kb(digits = 3)
```

design	target	msq_where_equal
d20	on	NA
d20	mid	NA
d20	off	NA
phq9	on	1.499
phq9	mid	1.562
phq9	off	1.491

Sum score

With complete data the WLE is a monotone function of the sum score, so their rank correlations with θ_1 should agree. Metric E differs between the two because the WLE stretches the scale at the extremes.

```
cell |>
  filter(rho %in% c(0.3, 0.5, 0.7, 0.9)) |>
  select(design, target, rho, spear_wle_a, spear_sum_a, E_wle, E_sum) |>
  kb(digits = 4)
```

design	target	rho	spear_wle_a	spear_sum_a	E_wle	E_sum
d20	on	0.3	0.9318	0.9318	0.0566	0.0507

design	target	rho	spear_wle_a	spear_sum_a	E_wle	E_sum
d20	on	0.5	0.9335	0.9335	0.0496	0.0457
d20	on	0.7	0.9353	0.9353	0.0398	0.0379
d20	on	0.9	0.9368	0.9368	0.0239	0.0235
d20	mid	0.3	0.9326	0.9326	0.0488	0.0433
d20	mid	0.5	0.9342	0.9342	0.0432	0.0392
d20	mid	0.7	0.9354	0.9354	0.0359	0.0330
d20	mid	0.9	0.9371	0.9371	0.0221	0.0210
d20	off	0.3	0.9349	0.9349	0.0330	0.0284
d20	off	0.5	0.9358	0.9358	0.0312	0.0267
d20	off	0.7	0.9369	0.9369	0.0258	0.0220
d20	off	0.9	0.9377	0.9377	0.0166	0.0142
phq9	on	0.3	0.9316	0.9316	0.0939	0.0969
phq9	on	0.5	0.9350	0.9350	0.0870	0.0881
phq9	on	0.7	0.9383	0.9383	0.0715	0.0729
phq9	on	0.9	0.9427	0.9427	0.0443	0.0452
phq9	mid	0.3	0.9297	0.9297	0.0952	0.0997
phq9	mid	0.5	0.9335	0.9335	0.0866	0.0902
phq9	mid	0.7	0.9374	0.9374	0.0719	0.0757
phq9	mid	0.9	0.9418	0.9418	0.0445	0.0476
phq9	off	0.3	0.9346	0.9346	0.0772	0.0776
phq9	off	0.5	0.9370	0.9370	0.0705	0.0702
phq9	off	0.7	0.9394	0.9394	0.0594	0.0594
phq9	off	0.9	0.9428	0.9428	0.0376	0.0376

Stability in n

On-target items only. If an impact metric is estimated without bias, its mean at n = 2,000 and 5,000 matches n = 20,000. The squared and absolute metrics (B, C, F) could be biased upward at small n by estimation noise that does not cancel between datasets (a) and (b).

```
cell_all |>
  filter(target == "on", rho %in% c(0.3, 0.5, 0.7, 0.85, 0.95, 1)) |>
  select(design, rho, n, msq, A_rmse_ratio, B_shift, B_gt1, C_kl, E_wle,
         F_sem) |>
  arrange(design, desc(rho), n) |>
  kb(digits = c(0, 2, 0, 3, 3, 3, 3, 4, 3, 3))
```

design	rho	n	msq	A_rmse_ratio	B_shift	B_gt1	C_kl	E_wle	F_sem
d20	1.00	2000	0.997	1.000	0.000	0.000	0.0000	0.000	0.000
d20	1.00	5000	1.001	1.000	0.000	0.000	0.0000	0.000	0.000
d20	1.00	20000	1.001	1.000	0.000	0.000	0.0000	0.000	0.000

design	rho	n	msq	A_rmse_ratio	B_shift	B_gt1	C_kl	E_wle	F_sem
d20	0.95	2000	1.046	1.000	0.041	0.000	0.0106	0.015	0.018
d20	0.95	5000	1.042	0.999	0.039	0.000	0.0103	0.017	0.018
d20	0.95	20000	1.042	0.999	0.039	0.000	0.0104	0.017	0.017
d20	0.85	2000	1.120	1.000	0.071	0.000	0.0181	0.029	0.050
d20	0.85	5000	1.131	0.998	0.071	0.000	0.0185	0.028	0.052
d20	0.85	20000	1.125	0.999	0.071	0.000	0.0185	0.029	0.052
d20	0.70	2000	1.243	1.000	0.106	0.000	0.0275	0.041	0.104
d20	0.70	5000	1.242	0.999	0.104	0.000	0.0272	0.040	0.096
d20	0.70	20000	1.246	1.000	0.104	0.000	0.0272	0.040	0.100
d20	0.50	2000	1.400	1.003	0.141	0.000	0.0378	0.050	0.173
d20	0.50	5000	1.396	1.004	0.140	0.000	0.0374	0.051	0.167
d20	0.50	20000	1.398	1.003	0.140	0.000	0.0374	0.050	0.171
d20	0.30	2000	1.546	1.013	0.172	0.000	0.0477	0.054	0.260
d20	0.30	5000	1.546	1.008	0.172	0.000	0.0474	0.057	0.246
d20	0.30	20000	1.542	1.010	0.171	0.000	0.0473	0.057	0.255
phq9	1.00	2000	0.999	1.000	0.000	0.000	0.0000	0.000	0.000
phq9	1.00	5000	1.000	1.000	0.000	0.000	0.0000	0.000	0.000
phq9	1.00	20000	1.000	1.000	0.000	0.000	0.0000	0.000	0.000
phq9	0.95	2000	1.074	1.001	0.089	0.001	0.0178	0.031	0.066
phq9	0.95	5000	1.077	0.999	0.091	0.001	0.0181	0.032	0.068
phq9	0.95	20000	1.076	1.000	0.090	0.001	0.0181	0.032	0.065
phq9	0.85	2000	1.218	1.006	0.170	0.002	0.0362	0.056	0.180
phq9	0.85	5000	1.222	1.006	0.171	0.002	0.0364	0.053	0.193
phq9	0.85	20000	1.220	1.005	0.170	0.002	0.0363	0.053	0.187
phq9	0.70	2000	1.418	1.029	0.257	0.009	0.0653	0.072	0.369
phq9	0.70	5000	1.416	1.032	0.256	0.009	0.0651	0.071	0.389
phq9	0.70	20000	1.411	1.033	0.255	0.008	0.0645	0.072	0.370
phq9	0.50	2000	1.637	1.086	0.346	0.030	0.1072	0.085	0.612
phq9	0.50	5000	1.645	1.089	0.349	0.032	0.1089	0.087	0.645
phq9	0.50	20000	1.646	1.088	0.348	0.031	0.1083	0.087	0.611
phq9	0.30	2000	1.850	1.163	0.427	0.066	0.1569	0.090	0.842
phq9	0.30	5000	1.863	1.162	0.430	0.067	0.1592	0.097	0.848
phq9	0.30	20000	1.854	1.162	0.428	0.065	0.1576	0.094	0.866

Overfit arm: the lower side, δ^-

Added 2026-10-12. Item i responds to θ_1 with discrimination $a \geq 1$ (GPCM) at the same location, against the same item with $a = 1$. Grid: a = 1, 1.15, 1.3, 1.5, 1.75, 2, 2.5 and 3. Same designs, targets, n = 20,000 and 40 replicates. Metric E is not defined, since there is no second dimension. Metric D, the true error SD over the reported SEM, is added for this arm only: overfit is classically said to make reported precision

wrong, and the other metrics cannot show that. D is shown as the ratio of (a) to (b), so 1 means the steep item does not change how honest the reported SEM is.

```
fs_s <- list.files("infit_impact_validation", pattern = "_slope[0-9]{3}[.]rds$",
                   full.names = TRUE)
Ls <- lapply(fs_s, readRDS)
Ss <- bind_rows(lapply(Ls, `[[`, "summary")) |>
  mutate(target = factor(target, levels = c("on", "mid", "off")))
Fs <- bind_rows(lapply(Ls, function(x) {
  f <- x$fsum; f <- f[f$n >= 200, ]
  d <- (f$bias_a - f$bias_b) / f$n; sem <- f$sem_b / f$n
  j <- which.max(abs(d) / sem)
  data.frame(design = x$base_info$design, target = x$base_info$target,
             slope = x$summary$rho[1], F_sem = abs(d[j]) / sem[j],
             F_theta = f$theta[j])
})) |>
  mutate(target = factor(target, levels = c("on", "mid", "off")))
cell_s <- Ss |>
  rename(slope = rho) |>
  group_by(design, target, rel_loc, slope) |>
  summarise(msq = mean(msq), other_min = mean(other_msq_min),
            other_max = mean(other_msq_max),
            A_rmse_ratio = mean(rmse_a / rmse_b),
            keep_vs_remove = mean(rmse_a / rmse_c),
            B_shift = mean(shift_mean), B_gt1 = mean(shift_gt1),
            C_kl = mean(kl_mean), D_ratio = mean(D_a / D_b), D_b = mean(D_b),
            reps = n(), .groups = "drop") |>
  left_join(Fs, by = c("design", "target", "slope"))
```

0 replicates failed.

```
cell_s |>
  select(design, target, slope, msq, other_min, other_max, A_rmse_ratio,
         B_shift, B_gt1, C_kl, D_ratio, F_sem, keep_vs_remove) |>
  arrange(design, target, slope) |>
  kb(digits = c(0, 0, 2, 3, 3, 3, 3, 3, 4, 4, 3, 3, 3))
```

design	target	slope	msq	other_min	other_max	A_rmse_ratio	B_shift	B_gt1	C_kl	D_ratio	F_sem	keep_vs_remove
d20	on	1.00	0.998	0.983	1.016	1.000	0.000	0.0000	0.0000	1.000	0.000	0.976
d20	on	1.15	0.941	0.989	1.019	1.001	0.016	0.0004	0.0042	0.998	0.023	0.976
d20	on	1.30	0.889	0.989	1.022	1.002	0.029	0.0007	0.0078	0.997	0.043	0.977
d20	on	1.50	0.833	0.992	1.026	1.003	0.045	0.0010	0.0119	0.997	0.064	0.979
d20	on	1.75	0.780	0.995	1.028	1.004	0.060	0.0011	0.0160	0.996	0.086	0.980
d20	on	2.00	0.738	0.997	1.031	1.005	0.072	0.0012	0.0191	0.996	0.101	0.981
d20	on	2.50	0.678	0.998	1.035	1.006	0.091	0.0013	0.0239	0.996	0.126	0.982
d20	on	3.00	0.643	1.002	1.037	1.007	0.104	0.0014	0.0273	0.996	0.144	0.983
d20	mid	1.00	0.999	0.985	1.014	1.000	0.000	0.0000	0.0000	1.000	0.000	0.977
d20	mid	1.15	0.942	0.988	1.020	1.001	0.019	0.0004	0.0041	0.998	0.032	0.978

design	target	slope	msq	other_min	other_max	A_rmse_ratio	B_shift	B_gt1	C_kl	D_ratio	F_slope	keep_vs_remove
d20	mid	1.30	0.893	0.989	1.022	1.002	0.035	0.0008	0.0077	0.998	0.060	0.979
d20	mid	1.50	0.839	0.991	1.025	1.003	0.051	0.0010	0.0113	0.997	0.090	0.980
d20	mid	1.75	0.787	0.994	1.031	1.005	0.069	0.0013	0.0153	0.997	0.122	0.982
d20	mid	2.00	0.747	0.993	1.032	1.006	0.081	0.0014	0.0182	0.997	0.147	0.983
d20	mid	2.50	0.692	0.994	1.037	1.008	0.099	0.0016	0.0228	0.997	0.182	0.985
d20	mid	3.00	0.658	0.995	1.039	1.009	0.111	0.0017	0.0259	0.997	0.210	0.986
d20	off	1.00	1.000	0.985	1.017	1.000	0.000	0.0000	0.0000	1.000	0.000	0.981
d20	off	1.15	0.949	0.987	1.018	1.002	0.021	0.0005	0.0037	0.998	0.052	0.983
d20	off	1.30	0.905	0.988	1.021	1.004	0.038	0.0009	0.0068	0.998	0.100	0.985
d20	off	1.50	0.859	0.987	1.024	1.006	0.054	0.0013	0.0101	0.998	0.150	0.987
d20	off	1.75	0.820	0.988	1.029	1.010	0.070	0.0028	0.0136	0.998	0.208	0.990
d20	off	2.00	0.786	0.989	1.033	1.011	0.080	0.0046	0.0165	0.997	0.256	0.992
d20	off	2.50	0.743	0.987	1.036	1.016	0.095	0.0058	0.0208	0.998	0.323	0.996
d20	off	3.00	0.716	0.988	1.040	1.020	0.103	0.0066	0.0236	1.000	0.361	1.001
phq9	on	1.00	1.001	0.985	1.014	1.000	0.000	0.0000	0.0000	1.000	0.000	0.953
phq9	on	1.15	0.933	0.997	1.023	1.002	0.039	0.0013	0.0067	0.994	0.069	0.954
phq9	on	1.30	0.877	1.002	1.031	1.005	0.070	0.0023	0.0131	0.992	0.134	0.957
phq9	on	1.50	0.815	1.009	1.041	1.011	0.105	0.0041	0.0210	0.991	0.210	0.962
phq9	on	1.75	0.756	1.018	1.048	1.018	0.140	0.0052	0.0296	0.992	0.289	0.969
phq9	on	2.00	0.709	1.022	1.053	1.024	0.166	0.0070	0.0368	0.994	0.354	0.975
phq9	on	2.50	0.644	1.030	1.061	1.035	0.203	0.0087	0.0484	0.997	0.456	0.986
phq9	on	3.00	0.603	1.035	1.064	1.041	0.226	0.0126	0.0575	1.000	0.539	0.992
phq9	mid	1.00	1.001	0.987	1.013	1.000	0.000	0.0000	0.0000	1.000	0.000	0.951
phq9	mid	1.15	0.930	0.996	1.023	1.002	0.035	0.0009	0.0062	0.996	0.060	0.953
phq9	mid	1.30	0.875	1.000	1.030	1.005	0.063	0.0016	0.0118	0.994	0.109	0.956
phq9	mid	1.50	0.810	1.008	1.037	1.009	0.093	0.0022	0.0183	0.993	0.165	0.960
phq9	mid	1.75	0.754	1.013	1.044	1.013	0.123	0.0030	0.0253	0.994	0.219	0.964
phq9	mid	2.00	0.708	1.019	1.050	1.017	0.145	0.0043	0.0307	0.995	0.263	0.968
phq9	mid	2.50	0.641	1.024	1.057	1.024	0.178	0.0056	0.0402	0.998	0.341	0.974
phq9	mid	3.00	0.595	1.027	1.064	1.028	0.201	0.0079	0.0480	0.999	0.384	0.979
phq9	off	1.00	1.002	0.986	1.014	1.000	0.000	0.0000	0.0000	1.000	0.000	0.954
phq9	off	1.15	0.935	0.994	1.022	1.002	0.037	0.0009	0.0059	0.995	0.083	0.957
phq9	off	1.30	0.878	1.001	1.029	1.004	0.066	0.0019	0.0113	0.992	0.154	0.960
phq9	off	1.50	0.818	1.005	1.037	1.009	0.098	0.0029	0.0180	0.990	0.243	0.963
phq9	off	1.75	0.761	1.013	1.044	1.016	0.126	0.0035	0.0249	0.991	0.332	0.970
phq9	off	2.00	0.715	1.013	1.048	1.021	0.148	0.0054	0.0310	0.992	0.422	0.974

design	target	slope	msq	other_min	other_max	A_rmse_ratio	B_shift	B_gt1	C_kl	D_ratio	F_slope	keep_vs_remove
phq9	off	2.50	0.651	1.017	1.057	1.031	0.179	0.0064	0.0407	0.995	0.555	0.984
phq9	off	3.00	0.608	1.019	1.061	1.036	0.198	0.0079	0.0479	0.996	0.643	0.990

`other_min` and `other_max` show the cascade from one steep item, which now pushes the other items toward *underfit*.

Calibrated δ^{-}. The departure 1 − MSQ at which impact reaches the tolerance, read along the grid. `>` means not reached.

```
tol_s <- tol |> filter(!grepl("^E", label))
invert_low <- function(msq, v, tau) {
  dev <- 1 - msq
  o <- order(dev); dev <- dev[o]; v <- v[o]
  j <- which(v >= tau)[1]
  if (is.na(j)) return(sprintf(">%.2f", max(dev)))
  if (j == 1) return(sprintf("%.2f", dev[1]))
  x <- dev[j - 1] + (tau - v[j - 1]) * (dev[j] - dev[j - 1]) / (v[j] - v[j - 1])
  sprintf("%.2f", x)
}
cell_s |>
  group_by(design, target, rel_loc) |>
  group_modify(function(d, key) {
    tibble(label = tol_s$label,
           delta = mapply(function(m, t) invert_low(d$msq, d[[m]], t),
                          tol_s$metric, tol_s$tau))
  }) |>
  ungroup() |>
  pivot_wider(names_from = label, values_from = delta) |>
  kb(digits = 2)
```

design	target	rel_loc	A: RMSE +2.5%	A: RMSE +5%	B: mean shift 0.25 SEM	B: 5% move > 1 SEM	C: KL 0.03	F: 0.25 SEM
d20	on	0.07	>0.36	>0.36	>0.36	>0.36	>0.36	>0.36
d20	mid	−0.98	>0.34	>0.34	>0.34	>0.34	>0.34	>0.34
d20	off	−2.06	>0.28	>0.28	>0.28	>0.28	>0.28	0.21
phq9	on	−0.56	0.29	>0.40	>0.40	>0.40	0.25	0.22
phq9	mid	−0.76	0.37	>0.41	>0.41	>0.41	0.29	0.28
phq9	off	−1.48	0.31	>0.39	>0.39	>0.39	0.28	0.19

Joint arm: three misfitting items

Added 2026-10-12. The three target items respond to one shared θ_2 with correlation ρ, as in Part A. The counterfactual has all three fitting, and dataset (c) removes all three. The question is how much larger the joint impact is than the impact of each item alone, which decides how lenient per-item bands are when several items misfit.

```
load_mech <- function(pattern) {
  fl <- list.files("infit_impact_validation", pattern = pattern, full.names = TRUE)
```

```
  lst <- lapply(fl, readRDS)
  list(L = lst, S = bind_rows(lapply(lst, `[[`, "summary")),
       F = bind_rows(lapply(lst, function(x) {
         f <- x$fsum; f <- f[f$n >= 200, ]
         d <- (f$bias_a - f$bias_b) / f$n; sem <- f$sem_b / f$n
         j <- which.max(abs(d) / sem)
         data.frame(design = x$base_info$design, target = x$base_info$target,
                    par = x$summary$rho[1], F_sem = abs(d[j]) / sem[j])
       })))
}
J <- load_mech("_jointrho[0-9]{3}[.]rds$")
cell_j <- J$S |>
  rename(par = rho) |>
  group_by(design, par) |>
  summarise(msq_1 = mean(msq_1), msq_2 = mean(msq_2), msq_3 = mean(msq_3),
            other_min = mean(other_msq_min), other_max = mean(other_msq_max),
            A_rmse_ratio = mean(rmse_a / rmse_b),
            keep_vs_remove = mean(rmse_a / rmse_c),
            B_shift = mean(shift_mean), C_kl = mean(kl_mean),
            E_wle = mean(E_wle), D_ratio = mean(D_a / D_b), .groups = "drop") |>
  left_join(J$F |> select(design, par, F_sem), by = c("design", "par"))
cell_j |>
  arrange(design, desc(par)) |>
  kb(digits = c(0, 2, 3, 3, 3, 3, 3, 3, 3, 3, 4, 3, 3, 3))
```

design	par	msq_1	msq_2	msq_3	other_min	other_max	A_rmse_ratio	keep_vs_remove	B_shift	C_kl	E_wle	D_ratio	F_sem
d20	1.00	1.002	1.001	0.999	0.985	1.015	1.000	0.931	0.0000	0.0000	0.000	1.000	0.000
d20	0.95	1.031	1.026	1.021	0.980	1.011	1.005	0.937	0.1010	0.0331	0.044	1.010	0.063
d20	0.90	1.063	1.054	1.042	0.975	1.006	1.010	0.941	0.1430	0.0494	0.061	1.020	0.121
d20	0.85	1.093	1.080	1.063	0.971	1.001	1.016	0.945	0.1780	0.0636	0.074	1.032	0.179
d20	0.80	1.121	1.105	1.084	0.965	0.998	1.023	0.952	0.2060	0.0760	0.084	1.044	0.246
d20	0.70	1.179	1.154	1.120	0.956	0.990	1.040	0.968	0.2560	0.0998	0.099	1.073	0.362
d20	0.60	1.231	1.200	1.156	0.947	0.982	1.057	0.984	0.2970	0.1208	0.111	1.102	0.474
d20	0.50	1.284	1.245	1.188	0.938	0.974	1.077	1.002	0.3350	0.1420	0.120	1.135	0.578
d20	0.30	1.385	1.336	1.251	0.921	0.959	1.124	1.046	0.4030	0.1836	0.129	1.212	0.817
phq9	1.00	1.002	1.001	0.999	0.987	1.013	1.000	0.844	0.0000	0.0000	0.000	1.000	0.000
phq9	0.95	1.027	1.031	1.021	0.976	1.000	1.030	0.868	0.2430	0.0808	0.090	1.048	0.185
phq9	0.90	1.048	1.056	1.034	0.965	0.989	1.068	0.902	0.3480	0.1338	0.121	1.104	0.366
phq9	0.85	1.070	1.080	1.056	0.955	0.978	1.110	0.937	0.4280	0.1832	0.144	1.168	0.528
phq9	0.80	1.092	1.105	1.072	0.945	0.969	1.155	0.976	0.4940	0.2311	0.159	1.237	0.688
phq9	0.70	1.125	1.148	1.106	0.925	0.950	1.249	1.057	0.6000	0.3214	0.184	1.382	0.985
phq9	0.60	1.162	1.187	1.133	0.907	0.935	1.349	1.139	0.6900	0.4109	0.198	1.540	1.237
phq9	0.50	1.194	1.225	1.162	0.890	0.918	1.453	1.226	0.7670	0.4976	0.207	1.713	1.491
phq9	0.30	1.254	1.295	1.213	0.856	0.891	1.652	1.393	0.9080	0.6742	0.211	2.071	1.918

0 replicates failed. `msq_1` to `msq_3` are the on, mid and off target items (for d20, the items near 0, −1 and −2 logits).

Joint against single. For each ρ, the joint impact next to the sum of the three single-item impacts from the main arm. A ratio above 1 means the items do more harm together than separately.

```
single_sum <- cell |>
  group_by(design, rho) |>
  summarise(A_single_sum = sum(A_rmse_ratio - 1), C_single_sum = sum(C_kl),
            E_single_sum = sum(E_wle), F_single_max = max(F_sem),
            .groups = "drop") |>
  rename(par = rho)
cell_j |>
  left_join(single_sum, by = c("design", "par")) |>
  filter(par < 1) |>
  transmute(design, rho = par, A_joint = A_rmse_ratio - 1, A_single_sum,
            C_joint = C_kl, C_single_sum, C_ratio = C_kl / C_single_sum,
            E_joint = E_wle, E_single_sum, F_joint = F_sem, F_single_max) |>
  arrange(design, desc(rho)) |>
  kb(digits = c(0, 2, 4, 4, 4, 4, 2, 3, 3, 3, 3))
```

design	rho	A_joint	A_single_sum	C_joint	C_single_sum	C_ratio	E_joint	E_single_sum	F_joint	F_single_max
d20	0.95	0.0047	−0.0013	0.0331	0.0281	1.18	0.044	0.045	0.063	0.034
d20	0.90	0.0097	−0.0025	0.0494	0.0399	1.24	0.061	0.063	0.121	0.067
d20	0.85	0.0157	−0.0027	0.0636	0.0491	1.29	0.074	0.075	0.179	0.096
d20	0.80	0.0225	−0.0020	0.0760	0.0573	1.32	0.084	0.086	0.246	0.132
d20	0.70	0.0399	−0.0009	0.0998	0.0718	1.39	0.099	0.102	0.362	0.184
d20	0.60	0.0571	0.0022	0.1208	0.0851	1.42	0.111	0.115	0.474	0.250
d20	0.50	0.0768	0.0074	0.1420	0.0981	1.45	0.120	0.124	0.578	0.315
d20	0.30	0.1238	0.0232	0.1836	0.1225	1.50	0.129	0.138	0.817	0.460
phq9	0.95	0.0296	−0.0010	0.0808	0.0528	1.53	0.090	0.091	0.185	0.085
phq9	0.90	0.0675	0.0039	0.1338	0.0814	1.64	0.121	0.126	0.366	0.166
phq9	0.85	0.1098	0.0175	0.1832	0.1088	1.68	0.144	0.152	0.528	0.256
phq9	0.80	0.1554	0.0367	0.2311	0.1373	1.68	0.159	0.171	0.688	0.328
phq9	0.70	0.2495	0.0967	0.3214	0.1955	1.64	0.184	0.203	0.985	0.479
phq9	0.60	0.3489	0.1722	0.4109	0.2597	1.58	0.198	0.226	1.237	0.633
phq9	0.50	0.4527	0.2678	0.4976	0.3282	1.52	0.207	0.244	1.491	0.778
phq9	0.30	0.6515	0.4919	0.6742	0.4791	1.41	0.211	0.266	1.918	1.055

Joint band. The mean MSQ of the three items at which the joint impact reaches the tolerance, minus 1.

```
cell_j |>
  mutate(msq = (msq_1 + msq_2 + msq_3) / 3) |>
  group_by(design) |>
  group_modify(function(d, key) {
    tibble(label = tol$label,
           delta = mapply(function(m, t) invert(d$msq, d[[m]], t),
```

```
                          tol$metric, tol$tau))
  }) |>
  ungroup() |>
  pivot_wider(names_from = label, values_from = delta) |>
  kb()
```

design	A: RMSE +2.5%	A: RMSE +5%	B: mean shift 0.25 SEM	B: 5% move > 1 SEM	C: KL 0.03	E: 0.05 SD per SD	E: 0.10 SD per SD	F: 0.25 SEM
d20	0.11	0.18	0.15	>0.32	0.02	0.04	0.16	0.11
phq9	0.02	0.04	0.03	>0.25	0.01	0.01	0.03	0.03

Local dependence arm: response dependence

Added 2026-10-12. Uniform response dependence as in the log-linear Rasch model: item j, the item nearest the target item i in location, responds to $\theta_1 + \lambda x_i$, so a higher score on item i makes item j easier. The counterfactual is the same pair without dependence ($\lambda = 0$), and dataset (c) removes item j. Grid: λ = 0, 0.25, 0.5, 0.75, 1, 1.5 and 2. Metric E is not defined. Metric D (true error over reported SEM) is included, since this is where reported precision may become too optimistic.

```
D_ld <- load_mech("_ldlambda[0-9]{3}[.]rds$")
cell_ld <- D_ld$S |>
  rename(par = rho) |>
  mutate(target = factor(target, levels = c("on", "mid", "off"))) |>
  group_by(design, target, item, ld_partner, par) |>
  summarise(msq_j = mean(msq), msq_i = mean(msq_partner),
            other_min = mean(other_msq_min), other_max = mean(other_msq_max),
            A_rmse_ratio = mean(rmse_a / rmse_b),
            keep_vs_remove = mean(rmse_a / rmse_c),
            B_shift = mean(shift_mean), B_gt1 = mean(shift_gt1),
            C_kl = mean(kl_mean), D_ratio = mean(D_a / D_b), .groups = "drop") |>
  left_join(D_ld$F |> mutate(target = factor(target, levels = c("on", "mid", "off")))
|>
              select(design, target, par, F_sem),
            by = c("design", "target", "par"))
cell_ld |>
  rename(lambda = par) |>
  arrange(design, target, lambda) |>
  kb(digits = c(0, 0, 0, 0, 2, 3, 3, 3, 3, 3, 3, 4, 4, 3, 3))
```

design	target	item	ld_partner	lambda	msq_j	msq_i	other_min	other_max	A_rmse_ratio	keep_vs_remove	B_shift	B_gt1	C_kl	D_ratio	F_sem
d20	on	11	12	0.00	1.001	0.999	0.985	1.014	1.000	0.976	0.000	0.0000	0.0000	1.000	0.000
d20	on	11	12	0.25	0.974	0.996	0.987	1.016	1.003	0.979	0.023	0.0002	0.0034	1.002	0.009
d20	on	11	12	0.50	0.949	0.990	0.989	1.019	1.005	0.981	0.044	0.0004	0.0066	1.003	0.017
d20	on	11	12	0.75	0.930	0.986	0.989	1.022	1.008	0.983	0.062	0.0004	0.0092	1.005	0.022
d20	on	11	12	1.00	0.909	0.983	0.990	1.022	1.010	0.986	0.078	0.0005	0.0116	1.007	0.029
d20	on	11	12	1.50	0.882	0.971	0.992	1.026	1.015	0.990	0.103	0.0007	0.0154	1.010	0.039
d20	on	11	12	2.00	0.862	0.965	0.994	1.026	1.019	0.994	0.122	0.0008	0.0181	1.013	0.045

design	target	held_partner	lambda	msq_rmse	msq_ratio	other_mono	other_max	A_rmse_ratio	vs_remove	B_shift	B_gt1	C_LD	D_ratio	F_sem
d20	mid	6	7	0.00	0.998	1.000	0.985	1.016	1.000	0.977	0.000	0.0000	0.0000	1.000	0.000
d20	mid	6	7	0.25	0.976	0.996	0.986	1.017	1.003	0.979	0.025	0.0001	0.0035	1.002	0.013
d20	mid	6	7	0.50	0.952	0.990	0.988	1.020	1.006	0.983	0.047	0.0001	0.0065	1.004	0.025
d20	mid	6	7	0.75	0.929	0.986	0.988	1.020	1.009	0.985	0.066	0.0002	0.0092	1.006	0.036
d20	mid	6	7	1.00	0.913	0.984	0.989	1.024	1.011	0.988	0.082	0.0002	0.0113	1.008	0.046
d20	mid	6	7	1.50	0.883	0.975	0.990	1.025	1.017	0.993	0.106	0.0002	0.0149	1.011	0.064
d20	mid	6	7	2.00	0.862	0.968	0.991	1.029	1.021	0.997	0.121	0.0003	0.0173	1.014	0.075
d20	off	1	2	0.00	0.998	1.002	0.984	1.016	1.000	0.980	0.000	0.0000	0.0000	1.000	0.000
d20	off	1	2	0.25	0.976	0.996	0.986	1.017	1.004	0.984	0.023	0.0000	0.0030	1.002	0.023
d20	off	1	2	0.50	0.958	0.994	0.987	1.018	1.007	0.987	0.042	0.0001	0.0056	1.004	0.047
d20	off	1	2	0.75	0.940	0.987	0.987	1.018	1.011	0.991	0.058	0.0001	0.0079	1.007	0.073
d20	off	1	2	1.00	0.923	0.986	0.988	1.019	1.015	0.995	0.070	0.0001	0.0098	1.009	0.099
d20	off	1	2	1.50	0.893	0.978	0.988	1.022	1.023	1.002	0.090	0.0001	0.0129	1.013	0.147
d20	off	1	2	2.00	0.871	0.976	0.987	1.024	1.030	1.009	0.103	0.0001	0.0151	1.018	0.189
phq9	on	1	5	0.00	1.002	1.001	0.987	1.012	1.000	0.952	0.000	0.0000	0.0000	1.000	0.000
phq9	on	1	5	0.25	0.918	0.982	1.001	1.027	1.016	0.967	0.130	0.0018	0.0173	1.007	0.056
phq9	on	1	5	0.50	0.867	0.971	1.009	1.038	1.027	0.978	0.180	0.0027	0.0263	1.014	0.091
phq9	on	1	5	0.75	0.843	0.963	1.013	1.043	1.034	0.985	0.197	0.0035	0.0346	1.018	0.107
phq9	on	1	5	1.00	0.830	0.961	1.011	1.046	1.041	0.990	0.199	0.0046	0.0406	1.023	0.114
phq9	on	1	5	1.50	0.835	0.956	1.009	1.045	1.046	0.995	0.236	0.0062	0.0523	1.028	0.105
phq9	on	1	5	2.00	0.852	0.954	1.006	1.043	1.050	0.999	0.263	0.0072	0.0616	1.031	0.077
phq9	mid	6	2	0.00	1.002	1.000	0.987	1.013	1.000	0.953	0.000	0.0000	0.0000	1.000	0.000
phq9	mid	6	2	0.25	0.897	0.976	1.002	1.031	1.018	0.969	0.139	0.0016	0.0179	1.009	0.073
phq9	mid	6	2	0.50	0.829	0.956	1.010	1.044	1.033	0.984	0.186	0.0025	0.0286	1.017	0.121
phq9	mid	6	2	0.75	0.790	0.942	1.010	1.050	1.042	0.992	0.208	0.0037	0.0398	1.024	0.142
phq9	mid	6	2	1.00	0.771	0.938	1.009	1.054	1.048	0.999	0.220	0.0047	0.0471	1.028	0.148
phq9	mid	6	2	1.50	0.760	0.927	1.007	1.056	1.053	1.003	0.250	0.0058	0.0567	1.032	0.139
phq9	mid	6	2	2.00	0.778	0.926	1.003	1.052	1.055	1.005	0.268	0.0069	0.0630	1.035	0.115
phq9	off	4	3	0.00	1.001	1.000	0.987	1.012	1.000	0.952	0.000	0.0000	0.0000	1.000	0.000
phq9	off	4	3	0.25	0.911	0.979	0.998	1.027	1.017	0.968	0.148	0.0012	0.0185	1.009	0.080
phq9	off	4	3	0.50	0.847	0.960	1.004	1.036	1.031	0.981	0.181	0.0018	0.0297	1.017	0.126
phq9	off	4	3	0.75	0.809	0.948	1.004	1.040	1.039	0.990	0.205	0.0024	0.0408	1.023	0.151
phq9	off	4	3	1.00	0.789	0.947	1.003	1.041	1.046	0.995	0.229	0.0032	0.0489	1.027	0.163
phq9	off	4	3	1.50	0.768	0.946	1.001	1.039	1.052	1.002	0.253	0.0047	0.0592	1.031	0.157
phq9	off	4	3	2.00	0.781	0.950	1.000	1.035	1.055	1.004	0.261	0.0057	0.0653	1.035	0.132

0 replicates failed. `item` is the target i and `ld_partner` the dependent item j. `msq_j` is the dependent item's MSQ and `msq_i` the target's. A `D_ratio` below 1 means the reported SEM understates the true error once the dependence is present.

Calibrated δ^- from response dependence, on the dependent item's MSQ, as 1 − MSQ.

```
tol_ld <- bind_rows(tol |> filter(!grepl("^E", label)),
                    tibble(metric = "D_gap", tau = 0.05,
                           label = "D: SEM 5% too optimistic"))
cell_ld |>
  mutate(D_gap = 1 - D_ratio) |>
  group_by(design, target) |>
  group_modify(function(d, key) {
    tibble(label = tol_ld$label,
           delta = mapply(function(m, t) invert_low(d$msq_j, d[[m]], t),
                          tol_ld$metric, tol_ld$tau))
  }) |>
  ungroup() |>
  pivot_wider(names_from = label, values_from = delta) |>
  kb()
```

design	target	A: RMSE +2.5%	A: RMSE +5%	B: mean shift 0.25 SEM	B: 5% move > 1 SEM	C: KL 0.03	F: 0.25 SEM	D: SEM 5% too optimistic
d20	on	>0.14	>0.14	>0.14	>0.14	>0.14	>0.14	>0.14
d20	mid	>0.14	>0.14	>0.14	>0.14	>0.14	>0.14	>0.14
d20	off	0.11	>0.13	>0.13	>0.13	>0.13	>0.13	>0.13
phq9	on	0.12	>0.17	0.15	>0.17	0.13	>0.17	>0.17
phq9	mid	0.13	0.22	0.22	>0.24	0.18	>0.24	>0.24
phq9	off	0.13	0.22	0.22	>0.23	0.15	>0.23	>0.23

Impact intervals

The floor turns a tolerance into a band and asks whether an item's MSQ interval lies inside it. The reverse is also possible: map the item's FWB interval for its MSQ through the impact curve, giving an interval for the item's *impact* in the metric's own units. No tolerance is needed to report it, and the user can judge the consequence directly.

The impact curve for an item combines the overfit arm (MSQ below 1, slope mechanism) and the main arm (MSQ above 1, second dimension), with impact 0 at MSQ = 1. Because impact is smallest at 1, an MSQ interval that contains 1 maps to an impact interval starting at 0. Otherwise the endpoints map to the endpoints.

The demonstration uses the FWB intervals from `fwb_msq_coverage.qmd` (d20 design, normal persons), mapped through the d20 curves calibrated here (resampled WLE persons, base SD 1.58 against 1.5, so the curves are close but not identical). The items are the one misfitting item at 0 logits ($\rho = .50$, population MSQ 1.38) and, for contrast, a fitting item under the null.

```
curve_for <- function(dsg, tgt, metric) {
  up <- cell |> filter(design == dsg, target == tgt) |>
    select(msq, v = all_of(metric))
  lo <- cell_s |> filter(design == dsg, target == tgt) |>
    select(msq, v = all_of(metric))
```

```
  bind_rows(lo, up, tibble(msq = 1, v = 0)) |>
    group_by(msq) |> summarise(v = mean(v), .groups = "drop") |>
    arrange(msq)
}
impact_interval <- function(cv, lo, hi) {
  g <- seq(lo, hi, length.out = 200)
  v <- approx(cv$msq, cv$v, xout = g, rule = 2)$y
  c(min(v), max(v))
}
fwb <- bind_rows(lapply(list.files("fwb_msq_coverage",
                                   pattern = "_n[0-9]+[.]rds$", full.names = TRUE),
                        readRDS))
demo <- bind_rows(
   fwb |> filter(cond == "xdim50_k1", item == 11) |> mutate(case = "misfitting,
0 logits"),
   fwb |> filter(cond == "null", item == 11) |> mutate(case = "fitting (null),
0 logits")
)
metrics_demo <- c(C_kl = "KL", F_sem = "F (SEM)", B_shift = "B (SEM)")
bind_rows(lapply(names(metrics_demo), function(m) {
  cv <- curve_for("d20", "on", m)
  demo |>
    rowwise() |>
    mutate(iv = list(impact_interval(cv, lo, hi)),
           true_v = approx(cv$msq, cv$v, xout = pop, rule = 2)$y) |>
    ungroup() |>
    mutate(metric = metrics_demo[[m]], low = vapply(iv, `[`, 1, 1),
           high = vapply(iv, `[`, 1, 2))
})) |>
  group_by(case, metric, n) |>
  summarise(msq_interval = sprintf("%.2f to %.2f", median(lo), median(hi)),
            impact_interval = sprintf("%.3f to %.3f", median(low), median(high)),
            true_impact = sprintf("%.3f", median(true_v)),
            coverage = 100 * mean(true_v >= low - 1e-9 & true_v <= high + 1e-9),
            .groups = "drop") |>
  arrange(case, metric, n) |>
  kb(digits = 1)
```

case	metric	n	msq_interval	impact_interval	true_impact	cover-age
fitting (null), 0 logits	B (SEM)	500	0.90 to 1.09	0.000 to 0.059	0.000	95.0
fitting (null), 0 logits	B (SEM)	2000	0.96 to 1.05	0.000 to 0.042	0.000	96.5
fitting (null), 0 logits	B (SEM)	5000	0.97 to 1.03	0.000 to 0.024	0.000	95.5
fitting (null), 0 logits	F (SEM)	500	0.90 to 1.09	0.000 to 0.051	0.000	95.0
fitting (null), 0 logits	F (SEM)	2000	0.96 to 1.05	0.000 to 0.025	0.000	96.5

case	metric	n	msq_interval	impact_interval	true_impact	cover-age
fitting (null), 0 logits	F (SEM)	5000	0.97 to 1.03	0.000 to 0.016	0.000	95.5
fitting (null), 0 logits	KL	500	0.90 to 1.09	0.000 to 0.015	0.000	95.0
fitting (null), 0 logits	KL	2000	0.96 to 1.05	0.000 to 0.011	0.000	96.5
fitting (null), 0 logits	KL	5000	0.97 to 1.03	0.000 to 0.006	0.000	95.5
misfitting, 0 logits	B (SEM)	500	1.25 to 1.50	0.105 to 0.161	0.135	94.5
misfitting, 0 logits	B (SEM)	2000	1.32 to 1.44	0.122 to 0.150	0.135	95.0
misfitting, 0 logits	B (SEM)	5000	1.34 to 1.42	0.126 to 0.144	0.135	90.5
misfitting, 0 logits	F (SEM)	500	1.25 to 1.50	0.101 to 0.228	0.160	94.5
misfitting, 0 logits	F (SEM)	2000	1.32 to 1.44	0.130 to 0.197	0.160	95.0
misfitting, 0 logits	F (SEM)	5000	1.34 to 1.42	0.140 to 0.182	0.160	90.5
misfitting, 0 logits	KL	500	1.25 to 1.50	0.027 to 0.044	0.036	94.5
misfitting, 0 logits	KL	2000	1.32 to 1.44	0.032 to 0.041	0.036	95.0
misfitting, 0 logits	KL	5000	1.34 to 1.42	0.034 to 0.039	0.036	90.5

The intervals are medians over the 200 datasets per cell, and coverage is the share of datasets whose impact interval contains the impact at the item's population MSQ. Because the mapping is monotone on each side of 1, coverage follows the coverage of the MSQ interval.

Relation to Part A

Part A used the d20 design with normal persons and three misfitting items. Its misfitting items had population MSQ 1.43 (0 logits), 1.37 (-1) and 1.27 (-2) at $\rho = .15$, 1.27, 1.23 and 1.17 at $\rho = .50$, and 1.14, 1.12 and 1.09 at $\rho = .75$. With one misfitting item at $\rho = .50$, the item at 0 logits had 1.38. Comparing these with the calibrated δ^{+} shows which of them the floor would flag at each tolerance.

The cascade-affected items in Part A had MSQ about 0.92, which is overfit. Whether the floor clears them depends on δ^{-}, the lower side of the band, which second-dimension misfit cannot calibrate. That check needs an overfit mechanism (for example a slope above 1) and is not part of this study.

Interpretation

Written 2026-10-12 after the run. 90 cells (54 at n = 20,000, 36 for the stability check), 3,600 replicates, none failed, a few minutes on 10 cores. At $\rho = 1$ every impact metric is exactly 0 and the MSQ is 1.00, as the counterfactual requires.

1. The calibration does not depend on n. For the on-target items, every metric at n = 2,000 and 5,000 agrees with n = 20,000 within Monte Carlo noise, at every ρ. Because (a) and (b) share all data except item i, estimation noise cancels and does not inflate the absolute or squared metrics. So the calibration can run at a moderate n.

2. In a 20-item test, one second-dimension item barely affects accuracy or individual scores. Even at $\rho = .3$ (MSQ about 1.5), RMSE against θ_1 rises by at most 1 percent, the mean score shift is 0.11

to 0.17 SEM, and almost nobody moves by more than 1 SEM. Tolerances on A and B are not reached within the grid. Metric A is nearly flat for d20, which is why its rank correlation with MSQ is only 0.6 to 0.7, and it cannot be inverted there.

3. The metrics that do respond in a long test are KL, group bias and conditional bias. For d20, KL reaches .03 at MSQ 1.29 to 1.39. The spurious group difference reaches 0.05 SD per SD of the item-specific factor only for the on-target item, at MSQ 1.41. The worst conditional bias reaches 0.25 SEM at MSQ 1.53 (on target), 1.40 (mid) and 1.26 (off target). KL responds before the mean shift does, because it squares the shift and is driven by the persons who move most.

4. Targeting runs in opposite directions for different metrics. The off-target item has the smallest MSQ and the smallest average impact (B, C, E), but the largest conditional bias (F): its misfit concentrates on the persons it is targeted at, in the tail. For a scale used against a cut-point in that tail, the off-target item is the one that matters.

5. A 9-item polytomous scale is far more sensitive. For phq9, every metric reaches its tolerance: RMSE +2.5 percent at MSQ 1.36 to 1.42, mean shift 0.25 SEM at 1.40 to 1.46, KL .03 at 1.17 to 1.20, group bias 0.05 SD at 1.20 to 1.29, and conditional bias 0.25 SEM at 1.22 to 1.28. The band is narrower because each item carries more of the score.

6. The calibrated bands depend on test length and on the tolerance, not mainly on the item. Within a design the three items give similar δ^+ for most metrics (for example KL .03: 0.29 to 0.39 for d20, 0.17 to 0.20 for phq9). Across designs and metrics δ^+ ranges from about 0.17 to more than 0.5. This is the consequence-based version of the Part A finding that no fixed band works everywhere.

7. Keeping a misfitting item is usually better than removing it. In d20 the RMSE with the item kept is 1.5 to 2.5 percent lower than with it removed at every ρ, down to .3. In phq9 keeping is better until MSQ about 1.5 (ρ about .6). This is more favourable to keeping than the linear-composite heuristic in the memory notes (crossover near $\rho = .72$ for 20 items), so that heuristic should not be relied on for IRT scoring.

8. The sum score behaves like the WLE. With complete data, the rank correlations with θ_1 are identical, as expected. The group bias in sum-score SD units is close to the WLE value, slightly smaller in d20.

9. One misfitting item already causes a cascade. In phq9 at $\rho = .3$ the other items fall to MSQ 0.87 to 0.89, on the overfit side. Whether the floor clears them depends on δ^-, from the overfit arm below.

Overfit arm (added 2026-10-12). 48 cells, 1,920 replicates, none failed.

10. A steep item does less harm than an equally large underfit. A slope of 3 brings the MSQ down to 0.64 to 0.72 in d20 and about 0.60 in phq9. In d20 none of the tolerances is reached within the grid (departures up to 0.28 to 0.36), except conditional bias for the off-target item (0.25 SEM at 1 − MSQ = 0.21). In phq9, KL reaches .03 at 0.25 to 0.29, RMSE +2.5 percent at 0.29 to 0.37, and conditional bias 0.25 SEM at 0.19 to 0.28. Compared with the upper side for phq9 (KL .03 at 0.17 to 0.20), the band is wider below 1 than above: roughly [0.72, 1.18] for KL .03.

11. Reported precision stays honest. The ratio of true error to reported SEM with the steep item, over the same with a fitting item (metric D), is 0.99 to 1.00 throughout. The claim that overfit makes the SEM or reliability too optimistic does not hold for a steep slope. It is more likely a property of redundancy between items (local dependence), which this arm does not generate.

12. A steep item costs a little accuracy, and keeping it still beats removing it. RMSE rises by up to 2 percent (d20) and 4 percent (phq9) at a slope of 3, because Rasch scoring gives the item the same weight as the others when it deserves more. Keeping it is still better than removing it in every cell except the d20 off-target item at a slope of 3, where the two are equal.

13. The cascade from a steep item goes the other way. One steep item pushes the other items' MSQ up, to at most 1.04 (d20) and 1.06 (phq9), well inside any upper band calibrated here.

Joint arm (added 2026-10-12). 18 cells, 720 replicates.

14. Misfitting items do more harm together than separately, and their MSQ shows less of it. With three items on a shared second dimension, KL is 1.2 to 1.5 times the sum of the three single-item KLs in d20, and 1.4 to 1.7 times in phq9. Conditional bias is well above the largest single-item value. Accuracy (A), flat for single items in d20, rises by up to 12 percent jointly, and by up to 65 percent in phq9. Group bias (E) is roughly additive. At the same time, each item's MSQ is *smaller* than when it misfits alone (d20 at $\rho = .3$: 1.39, 1.34 and 1.25 jointly, against 1.54, 1.52 and 1.43 alone), because the misfitting items now make up part of the total score that defines the expected responses.

15. So per-item bands fail when a coherent group of items misfits. The joint band on the items' mean MSQ is 0.01 to 0.04 in phq9 for every metric except the share moving more than 1 SEM (three of nine items on a second dimension), and 0.02 (KL) to 0.18 in d20. The impact is large while the MSQ hardly moves. This is not a calibration problem that a tighter band can fix: a subscale shows up in dimensionality diagnostics (residual PCA, Martin-Löf, Q3 clusters) better than in item fit. In the joint condition the reported SEM also becomes too optimistic as a measure of error in θ_1 (metric D up to 1.21 in d20 and 2.07 in phq9), because the scores measure a composite.

16. Removing a coherent misfitting group can pay off. In d20, keeping the three items is better than removing them down to ρ about .5. In phq9, removing them is better below ρ of about .75. At $\rho = .6$ the RMSE is 14 percent higher with the three items kept than with them removed.

Response dependence arm (added 2026-10-12). 42 cells, 1,680 replicates.

17. Response dependence produces overfit in the dependent item, with small consequences. At $\lambda = 2$ the dependent item's MSQ falls to 0.86 to 0.87 in d20 and 0.78 to 0.85 in phq9, and the target item's to 0.93 to 0.98. In d20 no tolerance is reached except RMSE +2.5 percent for the off-target pair. In phq9 KL reaches .03 at $1 - \mathrm{MSQ}$ = 0.13 to 0.18 and RMSE +2.5 percent at 0.12 to 0.13. Per unit of MSQ departure, response dependence harms more than a steep slope (phq9 KL .03 at 0.25 to 0.29 for the slope), so the lower band it implies is narrower, about 0.85 in phq9.

18. One dependent pair barely changes reported precision. Metric D is 1.01 to 1.02 in d20 and at most 1.04 in phq9, so the SEM is at most 4 percent too optimistic. The 5 percent tolerance is not reached. The optimism about precision that local dependence is known for needs more than one pair, or a stronger dependence.

19. MSQ is not monotone in the strength of dependence. In phq9 the dependent item's MSQ is lowest at λ around 1 to 1.5 and rises again at 2, while KL and the score shift keep growing. The inversion along the grid reads the first crossing, which is right for moderate dependence, but MSQ is a poor guide to strong dependence. Q3 or the log-linear dependence parameter itself would be.

Impact intervals.

20. The mapping works and reads naturally. For the misfitting item at 0 logits ($\rho = .50$, normal persons), the median interval at n = 2,000 is a KL of .032 to .041, a mean score shift of 0.12 to 0.15 SEM, and a worst conditional bias of 0.13 to 0.20 SEM, around the true values .036, 0.135 and 0.160. For a fitting item the intervals start at 0 and end at .011 (KL), 0.04 (B) and 0.03 (F) at n = 2,000. Coverage is 90.5 to 96.5 percent, inherited from the MSQ interval, since the mapping is monotone on each side of 1. This coverage does not test the impact curve itself, which was calibrated on resampled WLE persons while the FWB data used normal persons.

Relation to Part A. With KL .03 as the tolerance, the d20 bands are about 1.29 (0 logits), 1.31 (-1) and 1.39 (-2) on the upper side, and wider than 0.28 to 0.36 below 1 (the tolerance is not reached within the grid). Part A's misfitting items at $\rho = .15$ (1.43, 1.37, 1.27) would be flagged at 0 and −1 logits but not at −2, and its items at $\rho = .50$ and .75 would not be flagged at all. Its cascade-affected items, at MSQ about 0.92, lie far inside the lower band and would be cleared at any n large enough to make the interval narrow. So with impact-calibrated bands the cascade does not lead to flags. The calibration treats one misfitting item at a time, while Part A had three. Several misfitting items together have a larger joint

impact than each alone, so per-item bands are too lenient when several items misfit. A joint calibration, or recalibration after the worst item is dealt with, is needed for that case.

Limits. Four mechanisms (second dimension, steep slope, three items on a shared second dimension, and response dependence between one pair), two designs, and resampled WLE persons with a discrete distribution. The joint and dependence arms use one configuration each (three items, one pair). The tolerances are illustrations.